

Multiple Choice (10 marks)

1. A man is informed that, when a pair of dice were rolled, the resulting sum was a "seven". How much information is there in this message?
 - (a) 4 bits
 - (b) 1.5 bits
 - (c) 2.585 bits
 - (d) 1 bit
 - (e) 3.5 bits
2. Length of the cable is doubled, its capacitance C will be
 - (a) remains the same then doubled.
 - (b) unchanged.
 - (c) one-fourth.
 - (d) doubled.
 - (e) halved then doubled.
3. In an SR latch built from NOR gates, which condition is not allowed
 - (a) $S=0, R=2$
 - (b) $S=2, R=2$
 - (c) $S=1, R=1$
 - (d) $S=1, R=-1$
 - (e) $S=1, R=2$
4. Obtain the partial-fraction expansion of $F(s) = [(s + 2) / \{(s + 1)^2 (s + 3)\}]$
 - (a) $F(s) = [(1 / 5) / (s + 1)] + [(2 / 5) / (s + 1)^2] - [(2 / 5) / (s + 3)]$
 - (b) $F(s) = [(1 / 2) / (s + 1)] - [(1 / 2) / (s + 1)^2] + [(1 / 4) / (s + 3)]$
 - (c) $F(s) = [(3 / 4) / (s + 1)] - [(1 / 8) / (s + 1)^2] - [(5 / 8) / (s + 3)]$
 - (d) $F(s) = [(1 / 2) / (s + 1)^2] + [(1 / 4) / (s + 1)] - [(1 / 4) / (s + 3)]$
 - (e) $F(s) = [(1 / 6) / (s + 1)] + [(1 / 3) / (s + 1)^2] - [(1 / 2) / (s + 3)]$
5. Obtain the time function corresponding to $F(s) = [(s^2 + 3s + 1) / \{(s + 1)^3 (s + 2)^2\}]$
 - (a) $F(t) = - (3/2) t^2 e^{-t} + 5t e^{-t} - 6e^{-t} + (2/3) t e^{-2t} + 3e^{-2t}$
 - (b) $F(t) = (1/3) t^2 e^{-t} - t e^{-t} + e^{-t} - (2/3) t e^{-2t} - 3e^{-2t}$
 - (c) $F(t) = e^{-t} - 2t e^{-t} + (1/2) t^2 e^{-t} - e^{-2t} + (1/2) t e^{-2t}$
 - (d) $F(t) = - (1/2) t^2 e^{-t} + 3t e^{-t} - 4e^{-t} + t e^{-2t} + 4e^{-2t}$
 - (e) $F(t) = - (1/2) t^2 e^{-t} + 3t e^{-t} - 4e^{-t} - t e^{-2t} - 4e^{-2t}$

True / False (10 marks)

Answer True or False

6. True or False: A 3-input NAND gate will have 6 entries in its truth table, which is the number of inputs.
7. True or False: The evaluated vector function at the point $(-1, 1, 2)$ results in $-2 + e^1$.
8. True or False: The energy dissipated in a 2 resistor with current $i(t) = \cos^2(t)$ from $t = 0$ to $t = 5$ s is 5 J.
9. True or False: A TV signal with 80% audio modulation has a carrier swing of 12 kHz and a frequency deviation of 24 kHz.
10. True or False: Resolution refers to the smallest change in a measured variable that an instrument can detect.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Given a cylinder with a volume of 10 ft³ and a safe working pressure of 500 psig, analyze the factors that determine the maximum operating temperature for 1 lb-mole of propane it contains, and explain your reasoning.
12. Discuss the feasibility of designing a quarter-wave transformer to match a 20-ohm source impedance to a $150 + j40$ -ohm load impedance at 50MHz, including limitations of this approach.

End of Paper